Fresh native cutoff, typed-entry and solver comparison

Native F4/F5, F6 and both root-solving families; bounded public Koblitz m2 Boolean fixtures. Final same-host table: run 20261019-024352-boolean-native-policy-ty-fb16d2. Five admitted samples per arm, four primary fixtures, fresh paired planted holdouts231/232. Complete roots and first-witness decomposition have separate boundaries. Every output is checked against complete scalar roots or the original group equation.

Final host: Intel(R) Xeon(R) Platinum 8488C; rustc 1.99.1 (58cf8f9c3 2026-10-13); reserved CPUs 3,7; c7i.2xlarge on-demand, one Rayon worker, exclusive reserved physical core, isolation receipts and A/A. VM neighbours/frequency remain outside the reservation. Failed and contended attempts are retained, excluded, and counted in each JSON summary.

n17: icv1-f2m17-tm101-00378d4e,16 Boolean variables,205 base points, group order 131,174. n23: icv1-f2m23-tm5197-1f85e9e1,22 Boolean variables,2071 base points, group order 8,393,806. Exact constructor output is fixture-identities.json. n19 is a 36-variable implementation-limit refusal, not an intermediate-size timing.

Planted target uses seed230; generator control is the registered generator. The old driver flag random=true does not generate a random target for its single-target invocation. TARGET_LABEL_CORRECTION.md explicitly corrects that earlier label without rewriting frozen inputs or raw records. These cases do not establish an average-case ranking.

Complete-root runs start from the prepared ANF or JSON representation for that API; system construction and JSON encoding are charged when rebuilt in the decomposition wrapper. Cold decomposition charges fresh target-dependent ANF construction, solving, lifting and verification; Semaev arms additionally charge fresh oracle/table preparation once. Common curve/base/context construction is outside every solver interval. Independent complete-root and final group-audit checks run after elapsed capture for all arms. Instrumented query timings include common driver metadata/wrapper overhead and are not bare-kernel latency. No whole-IC/rho ratio is supplied.

F6 is compared through the repository's first-witness decomposition APIs; no complete-original-root F6 API is benchmarked here. The original n23 planted system has two Boolean roots and the generator system has none. Exact complete-root output is a separate obligation from returning one group witness.

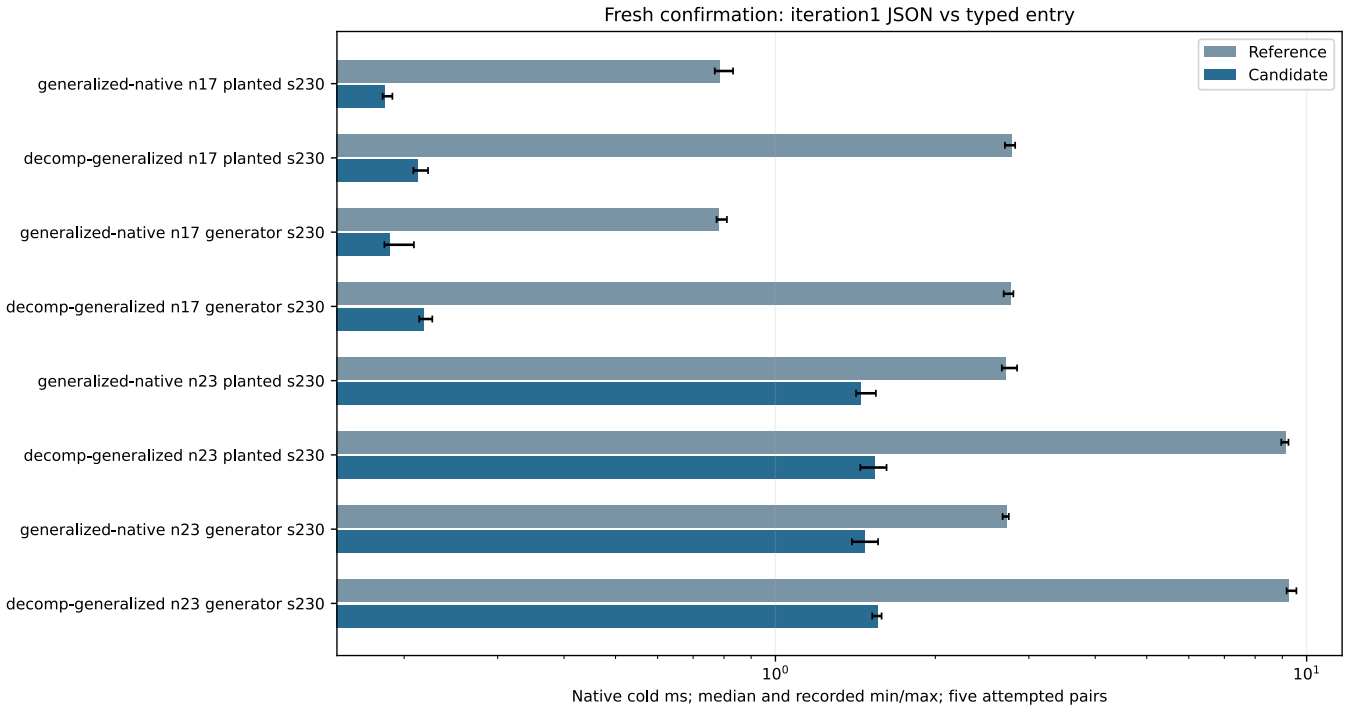
Iteration1 generalized JSON improvements borrow input, simplify exact evaluation and integrate the existing SIMD Gray kernel. Same-candidate ablations found SIMD slower at 16 variables and faster at 22; auto now keeps scalar below 22. Fresh confirmation measures that policy and the typed F2BoolPoly entry, which avoids JSON/BigUint conversion while retaining exact counts, output caps, budgets, original coordinates and root verification. This is engineering integration, not a new enumeration algorithm.

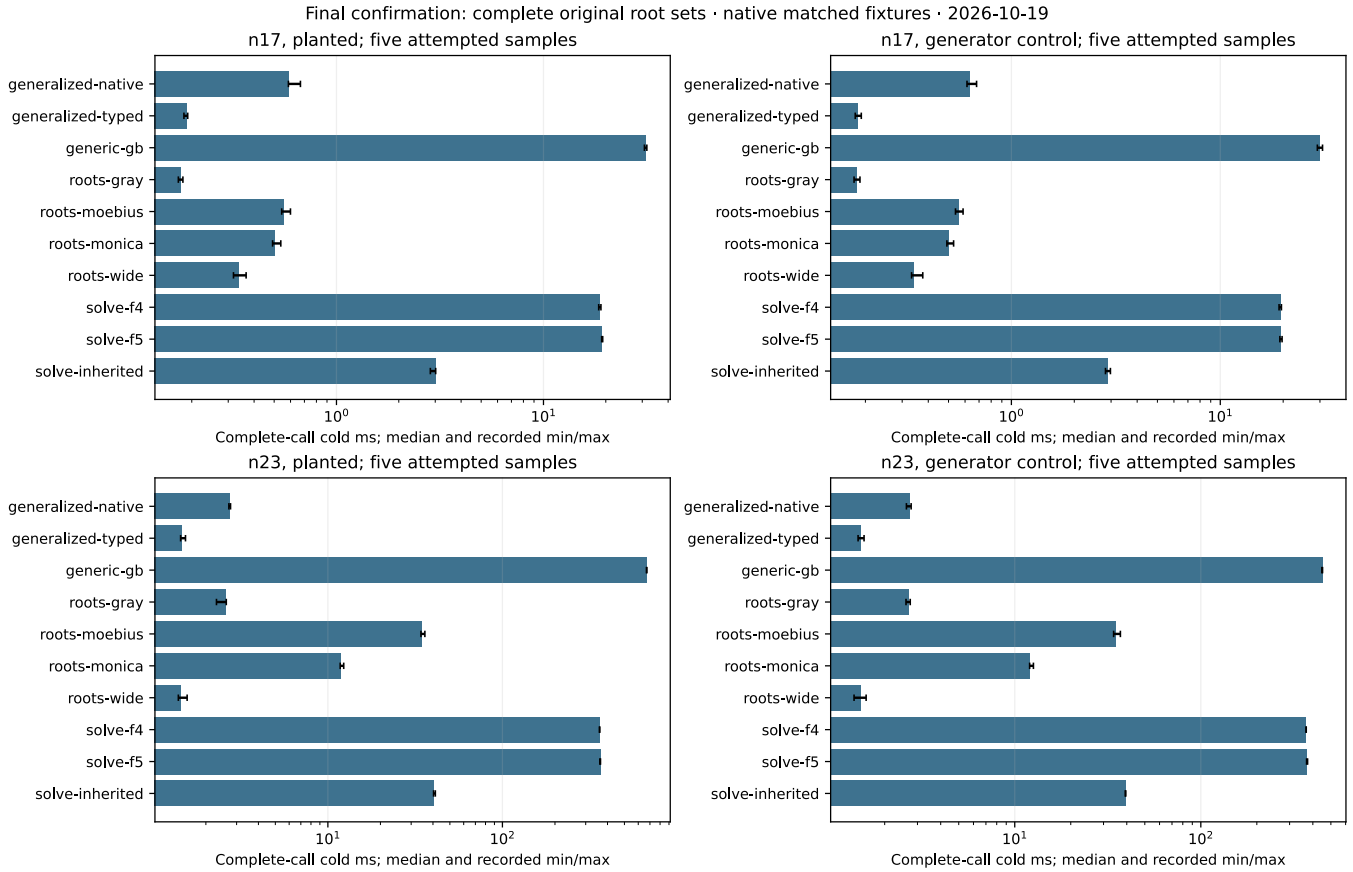
Single-write, SIMD tables, serial row clearing and batched Semaev setup remain opt-in; extra SIMD tables were neutral. The first serial guard regressed one four-worker control; the refined guard requires explicit RAYON_NUM_THREADS=1 and a current one-worker pool. Generalized auto uses scalar below22 variables and wide Gray when available at22 or more. Typed input avoids JSON; exact counts and original-root verification remain enforced. The18-arm costs precede serial-clear experiments and do not claim their complete-root speedup.

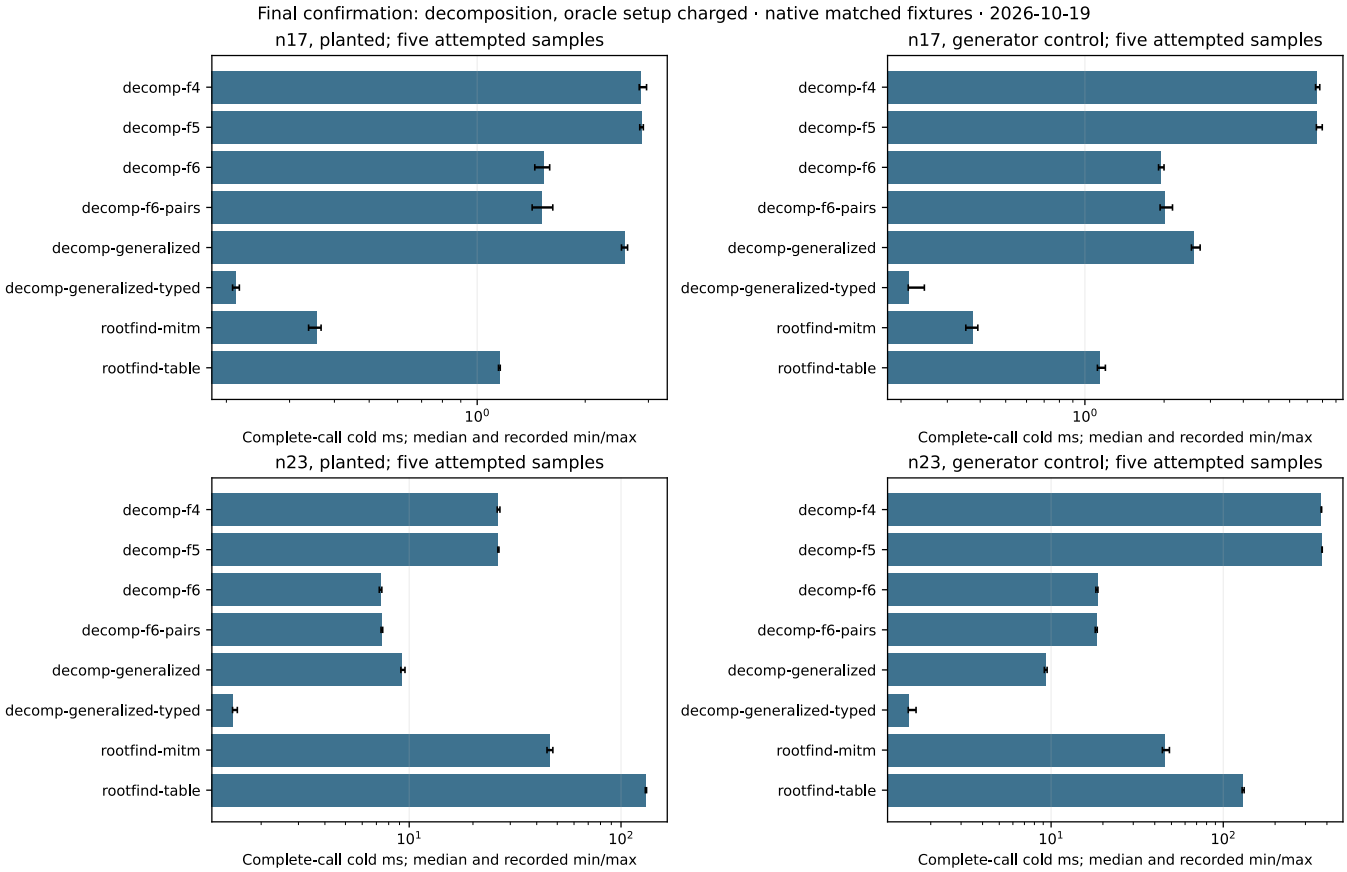
Fresh confirmation retains all outcomes: all 16 typed cells qualify faster than iteration1; n17 prepared JSON roots qualify faster on all four primary/holdout cases. n23 prepared JSON controls are neutral. One n23 planted JSON decomposition control regresses about 3.5% (before/after 0.9654); it is not hidden or described as regression-free. The typed route qualifies about 6x against that frozen wrapper on the same fixture.

Native comparison/admission, statistics and ratios are computed by the Rust analyzers. Five paired ratios use all 3125 bootstrap resamplings with percentile indices78/3046; qualification additionally requires effect greater than max(2%, twice median absolute A/A noise). Charts show recorded medians and min/max; full CI, individual rounds and rejected attempts remain in JSON/raw archive. Python only renders these prepared values.

Evidence: initial-summary.json and PROTOCOL.md (run 014036 baseline); optimization-summary.json and OPTIMIZATION_PROTOCOL.md (run 013858 first candidate); final-summary.json and FINAL_PROTOCOL.md (run 014050 Semaev); policy-summary.json, policy-comparison-summary.json and POLICY_PROTOCOL.md (run 024352 confirmation). Source/binary hashes, protocols, commands, every retry, preparation failures and corrected/failed auxiliary Callgrind profiles are retained. Callgrind Ir is not native runtime. See RAW_EVIDENCE.md for archive extraction and immutable job locations.


Fresh confirmation: iteration1 JSON vs typed entry
Reference
Candidate
generalized-native n17 planted s230
decomp-generalized n17 planted s230
generalized-native n17 generator s230
decomp-generalized n17 generator s230
generalized-native n23 planted s230
decomp-generalized n23 planted s230
generalized-native n23 generator s230
decomp-generalized n23 generator s230
10^0
10^1
Native cold ms; median and recorded min/max; five attempted pairs


Final confirmation: complete original root sets · native matched fixtures · 2026-10-19
n17, planted; five attempted samples
n17, generator control; five attempted samples
n23, planted; five attempted samples
n23, generator control; five attempted samples
generalized-native
generalized-typed
generic-gb
roots-gray
roots-moebius
roots-monica
roots-wide
solve-f4
solve-f5
solve-inherited
Complete-call cold ms; median and recorded min/max


Final confirmation: decomposition, oracle setup charged · native matched fixtures · 2026-10-19
n17, planted; five attempted samples
n17, generator control; five attempted samples
n23, planted; five attempted samples
n23, generator control; five attempted samples
decomp-f4
decomp-f5
decomp-f6
decomp-f6-pairs
decomp-generalized
decomp-generalized-typed
rootfind-mitm
rootfind-table
Complete-call cold ms; median and recorded min/max

n23 complete_roots, planted; five samples, ratio relative to F4

Method	Cold ms	n	F4 / method
generalized-native	2.7533	5	131.150
generalized-typed	1.4494	5	248.149
generic-gb	669.8408	5	0.538
roots-gray	2.5987	5	138.912
roots-moebius	34.4747	5	10.448
roots-monica	11.9093	5	30.193
roots-wide	1.4275	5	252.955
solve-f4	360.1913	5	1.000
solve-f5	363.9447	5	0.991
solve-inherited	40.4737	5	8.896

n23 complete_roots, generator; five samples, ratio relative to F4

Method	Cold ms	n	F4 / method
generalized-native	2.7096	5	135.873
generalized-typed	1.4774	5	249.159
generic-gb	449.6532	5	0.818
roots-gray	2.7031	5	136.175
roots-moebius	34.8805	5	10.553
roots-monica	12.0878	5	30.491
roots-wide	1.4900	5	247.049
solve-f4	368.1564	5	1.000
solve-f5	372.7797	5	0.988
solve-inherited	39.3968	5	9.342

n23 decomposition, planted; five samples, ratio relative to F4

Method	Cold ms	n	F4 / method
decomp-f4	26.3691	5	1.000
decomp-f5	26.2727	5	1.004
decomp-f6	7.3539	5	3.610
decomp-f6-pairs	7.3983	5	3.552
decomp-generalized	9.2244	5	2.859
decomp-generalized-typed	1.4781	5	17.840
rootfind-mitm	46.3986	5	0.564
rootfind-table	131.0207	5	0.201

n23 decomposition, generator; five samples, ratio relative to F4

Method	Cold ms	n	F4 / method
decomp-f4	370.4942	5	1.000
decomp-f5	375.1102	5	0.988
decomp-f6	18.6067	5	19.910
decomp-f6-pairs	18.3590	5	20.178
decomp-generalized	9.3417	5	39.691
decomp-generalized-typed	1.4931	5	248.331
rootfind-mitm	45.6419	5	8.117
rootfind-table	130.1315	5	2.853